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EVOLUTION OF A SECTION OF THE AFRICA–EUROPE PLATE BOUNDARY: PALEOMAGNETIC AND VOLCANOLOGICAL EVIDENCE FROM SICILY

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New paleomagnetic data relative to Upper Cretaceous, Neogene and Quaternary volcanic rocks from eastern Sicily definitively indicate that Sicily is a part of the African plate, which collided with the European continental plate in Middle Miocene times. These data and the tectonic evolution of Sicily as inferred from the nature, age and distribution of volcanic products, are broadly consistent with the motions of Africa relative to Europe since the Upper Trias. During the Mesozoic, eastern Sicily was affected by extensional tectonics with associated alkali basaltic volcanism, and oceanic crust was produced in the meantime between the diverging African and European plates. Near the end of Mesozoic times the two plates started to converge with consequent consumption of oceanic crust. Different times of oceanic plate consumption along the Sicily–Calabria section of the plate boundary are suggested by the occurence of andesitic volcanism of different ages. The tectonic significance of late Tertiary to present basaltic activity in eastern Sicily is also discussed.

1. Introduction

Recent interpretations of the geophysics and geology of the central Mediterranean Sea in terms of the plate tectonic theory have indicated that the boundary between the African and European plates probably crosses Sicily [1]. Furthermore, it has been shown [2] that an island-arc structure exists in the

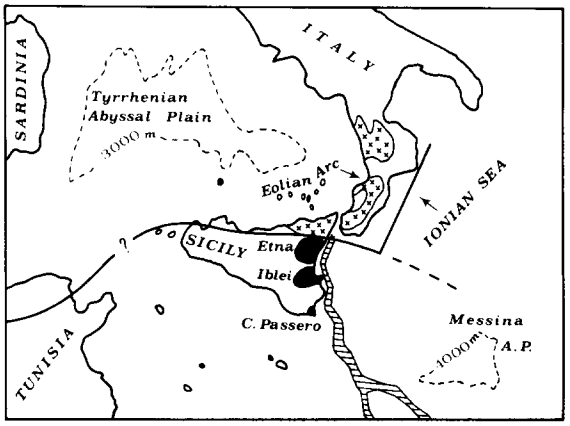


Fig. 1. Sketch map of Sicily in its Mediterranean setting. Continuous line indicates probable boundary between African and European plates [1, 2]. Crosses: Calabria-Peloritani crystalline belt. Black areas: basaltic volcanism in eastern Sicily. Dashed area: distensive fault scarp in western Ionian Sea [24].

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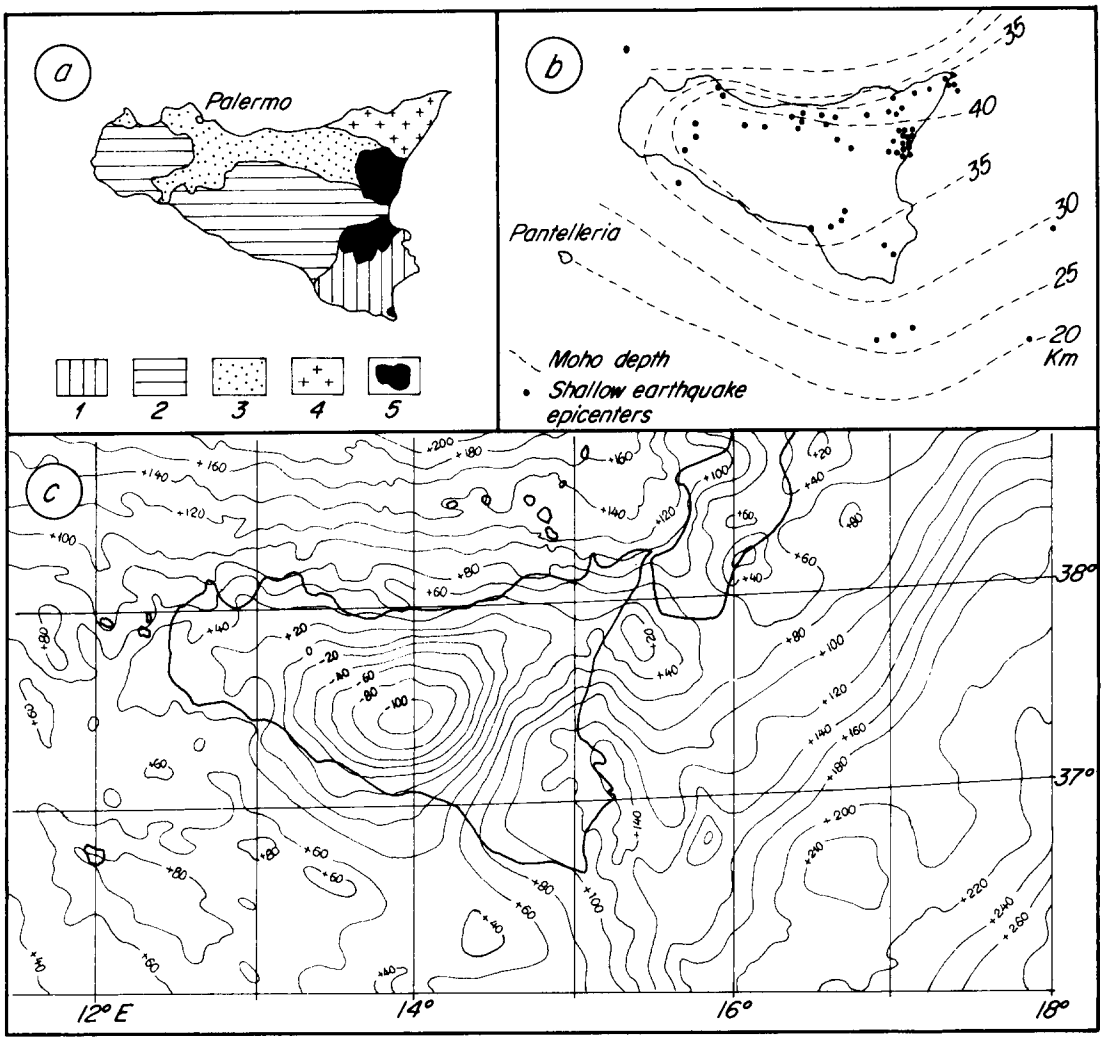


Fig. 2. (a) Geological sketch map of Sicily. 1 = Ragusa-Iblei carbonate platform; 2 = Neogene sedimentary basins; 3 = different nappe units (including Tusa flysch) 4 = Peloritani crystalline domain; 5 = Upper Cretaceous, Tertiary and Quaternary volcanic rocks.

(b) Distribution of shallow earthquakes in and around Sicily in the period 1893–1968 [10] and Moho depth contour lines [25, 6].

(c) Bouguer gravity map [25] (ρ = 2.67 g/cm^3).

southern Tyrrhenian Sea (Fig. 1). It includes a sediment filled trench along the Ionian coast of Calabria, a metamorphic belt (Calabrian–Peloritani), a calc-alkaline volcanic arc in an advanced stage of evolution (Eolian Islands) [2] and a backarc marginal basin (Tyrrhenian Abyssal Plain). The WNW-dipping seismic plane has a limited lateral extent of about 200 km [3, 4]. No evidence for an active Benioff zone exists along the plate boundary crossing northern Sicily.

The reconstruction of past movements of Sicily relative to Africa and the stable Europe is essential to understand how the present structural setting was attained. This reconstruction is attempted on the basis of paleomagnetic data and tectonic evolution as inferred from the distribution, age and nature of volcanism in eastern Sicily.

2. Geological and geophysical outline of Sicily

The main geophysical and geological features of Sicily are summarized in Fig. 2. Four main structural units can be recognized in Sicily [5]. The most external unit is represented by the Ragusa–Iblei carbonate platform. It is not affected by Alpine folding and appears to be a prolongation of the Saharan platform. This is supported by the 20 km thick continental-type crust evidenced by deep seismic sounding (DSS) in the Sicily Channel [6]. A Mio–Pliocene resedimentation basin (Caltanisetta basin) occurs in central Sicily. It is separated from the Ragusa carbonate platform by steep vertical faults. The basin subsided in the Upper Miocene and is filled with about 3 km of sediments. Since Lower Pliocene times Sicily has been affected by uprising. Upper Pliocene–Lower Quaternary deposits are found at present up to several hundred meters above sea level [7].

Bouguer gravity anomalies are greatly affected by the occurrence of this thick sedimentary basin (Fig. 2). The steep gradients on the north and east margins correspond to faults with relevant vertical throw.

A sequence of flysch nappes, ranging in age from Cretaceous to Miocene, occurs in North Sicily. One of these flysch units (Tusa flysch, Upper Oligocene–Lower Miocene) contains layers of tuffites and fragmented volcanic materials of calc-alkaline nature [8]. The uppermost unit is represented by the Peloritani crystalline massif, which is the Sicilian continuation of the Calabrian metamorphic belt.

Two DSS profiles are available for Sicily, one E–W across North Sicily [9], and the other from Palermo to Pantelleria (NNE–SSW) [6]. They indicate a typical continental crust with low-velocity layers. The crustal thickness increases from the Sicily Channel to the northern edge of Sicily (about 40 km) and then sharply decreases again when entering the Tyrrhenian Sea. The crustal thickness remains nearly constant along the northern part of Sicily (~40 km).

The Moho discontinuity is sharp underneath the Sicily Channel, whereas it is poorly defined within Sicily, suggesting a gradual crust–mantle transition.

The persistence of intensive tectonic activity within and around Sicily is testified by the numerous earthquakes (see [10]). Earthquakes in Sicily are generally shallow (<50 km) and mostly concentrated around the Caltanisetta basin. Shallow earthquakes also occur in the Ionian Sea, southeast of Sicily. Intermediate and deep earthquakes occur north of the island in relation to the WNW-dipping subduction zone. Focal solutions of shallow earthquakes are

TABLE 1
K–Ar dates of Upper Cretaceous dykes in eastern Sicily*

Sample No.	Locality	Coordinate	K%	^{40}Ar % rad	^{40}Ar rad (10^{-10} moles/g)	K–Ar age (m.y.)	Rock type
Ib 4	Capo Passero	15°07′ E 36°41′ N	1.09 (±0.02)	78.3	1.53	77.6 (±2.5)	alkaline basalt
Ib 5	Capo Passero	15°07′ E 36°41′ N	1.52 (±0.02)	70.4	1.95	70.7 (±2.0)	alkaline basalt
Ib 6	Capo Passero	15°07′ E 36°41′ N	0.90 (±0.01)	64.6	1.38	84.1 (±2.4)	alkaline basalt
Ib 31	Capo Passero	15°07′ E 36°41′ N	0.89 (±0.01)	24.0	1.27	78.3 (±2.7)	alkaline basalt
Ib 22	Mt. Tauro	15°12′52″ E 37°15′46″ N	1.19 (±0.01)	70.4	1.76	81.1 (±2.5)	alkaline basalt
Ib 23	Mt. Tauro	15°12′52″ E 37°15′46″ N	1.24 (±0.02)	66.2	1.77	78.5 (±2.5)	alkaline basalt
Ib 047	S. Cusmano	15°14′01″ E 37°11′28″ N	0.73 (±0.01)	52.3	1.02	76.6 (±2.4)	alkaline basalt

* $\lambda_e = 0.585 \times 10^{-10}\ yr^{-1}$; $\lambda_\beta = 4.72 \times 10^{-10}\ yr^{-1}$; $^{40}K/K = 1.22 \times 10^{-4}$ g/g.

TABLE 2
K–Ar ages of Mt. Iblei lavas*

Sample No.	Coordinates	K%	$^{40}Ar\%$ rad	^{40}Ar rad (10^{-12} moles/g)	K–Ar age (m.y.)	Magnetic polarity	Rock type
Ib 8	14°50'41" E 37°14'20" N	0.75 (±0.01)	7.7	7.16	5.36 (±0.37)	R	alkaline basalt
Ib 14	14°48'19" E 37°11'40" N	0.106 (±0.004)	3.5	0.577	3.05 (±0.47)	R	tholeiitic basalt
Ib 20	14°50'30" E 37°09'32" N	1.41 (±0.02)	18.3	7.67	3.05 (±0.13)	N	alkaline basalt
Ib 18	14°43'57" E 37°14'50" N	0.98 (±0.01)	8.3	4.99	2.85 (±0.14)	R	alkaline basalt
Ib 11	14°45'33" E 37°10'02" N	0.92 (±0.02)	11.5	3.88	2.40 (±0.13)	N	alkaline basalt
Ib 2	14°50'55" E 37°13'02" N	0.141 (±0.002)	4.7	0.582	2.40 (±0.20)	R	tholeiitic basalt
Ib 1	14°50'55" E 37°13'02" N	0.117 (±0.001)	1.7	0.459	2.20 (±0.60)	R	tholeiitic basalt
Ib 13	14°47'17" E 37°11'21" N	0.91 (±0.02)	34.1	3.50	2.16 (±0.07)	N	alkaline basalt
Ib 3	14°45'34" E 37°10'20" N	0.86 (±0.02)	9.4	2.95	1.75 (±0.10)	N	alkaline basalt
Ib 17	14°44'19" E 37°17'43" N	0.402 (±0.005)	3.0	1.21	1.71 (±0.30)	R	alkaline basalt

* $\lambda_e = 0.585 \times 10^{-10}\ yr^{-1}$; $\lambda_\rho = 4.72 \times 10^{-10}\ yr^{-1}$; $^{40}K/K = 1.22 \times 10^{-4}$ g/g.

available only for three earthquakes which occurred in 1968 in Valle del Belice (West Sicily). McKenzie [1] presented long-period solutions of these quakes, which are consistent with a thrust mechanism on E–W planes. This mechanism is the same as that observed by McKenzie [1] for earthquakes along the Gibraltar–Morocco–Algeria line, of which the North Sicily seismic line seems to be a prolongation.

No focal solutions are available for shallow earthquakes in eastern Sicily and the neighbouring Ionian Sea.

3. Volcanism

Eastern Sicily has been the site of volcanism of dominantly basaltic nature, since the Middle Trias. With the exception of the most recent episodes of Mt. Iblei and Mt. Etna, this volcanism occurred in a submarine environment giving rise to pillow-lavas and hyaloclastites emitted through fissures cutting a carbonate platform.

Only Creataceous or younger volcanics are found in outcrop, older products having been penetrated only in drillings in this area [11].

K–Ar ages are reported (Table 1) for dykes cutting hyaloclastites of three different Cretaceous outcrops (Capo Passero, St. Cusmano and Mt. Tauro; the latter are located at the eastern limit of Mt. Iblei). Ages of the Capo Passero samples are scattered between 71 and 81 m.y. When $^{40}Ar_{rad}$ contents of these samples are plotted against K constents, an isochron with positive intercept is obtained. This suggests (see [12]) that extraneous initial $^{40}Ar_{rad}$ is present in some phase. The most reliable age is, therefore, 71 m.y., being consistent within the errors with the age defined by the isochron. This age is also consistent with the stratigraphic evidence indicating an Upper Cretaceous age [13].

K–Ar ages of the Mt. Tauro samples are consistent

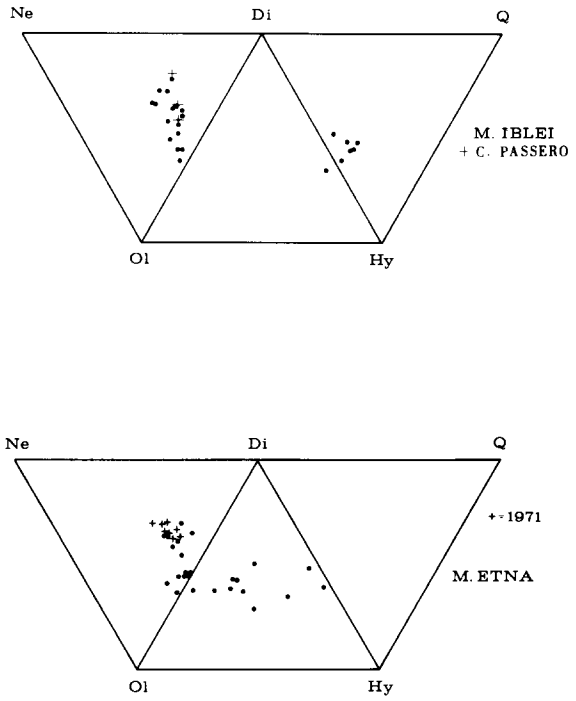


Fig. 3. Plot of normative mineralogy of Mt. Iblei and Mt. Etna basalts in basaltic tetrahedron. Norms were calculated assuming a constant Fe_2O_3 content of 1.5%. Upper Cretaceous samples from Capo Passero and Mt. Tauro (crosses) are also included in Mt. Iblei plot. Mt. Etna 1971 lavas are indicated by crosses. Data sources [26].

around 80 m.y., being also coherent with the reversed magnetic polarity of the formation.

From Cretaceous to Upper Miocene no episodes of basaltic volcanism are so far known in eastern Sicily. Some Eocene basic lavas are reported in western Sicily ([14]; Villari, unpublished thesis). It is noteworthy that a calc-alkaline volcanism occurred during Upper Oligocene–Early Miocene times, as evidenced by the andesitic nature of the volcanic fragments included in the Tusa formation [8].

Important basaltic activity has again occurred between Upper Miocene times and the present, mainly in the Mt. Iblei region and at Mt. Etna. The Iblei volcanic complex was built by a dominantly fissural activity in both shallow submarine and subaerial environments on a dominant northwest fracture system [15]. K–Ar ages ranging from 5.4 to 1.7 m.y. have been found for some subaerial lavas (Table 2), the results are consistent with paleomagnetic stratigraphy. The Etna volcano has grown, probably since the Sicilian stage [16] on a submarine base with dominant feeding fractures of ENE and NNW trends.

The volcanism of eastern Sicily has a dominantly alkaline character. The most basic products range from alkali basalts to basanites and, mainly at Mt. Etna, they are found associated with their differentiation products (hawaiites, mugearites).

Tholeiitic lavas are found among the Plio–Pleistocene products of Mt. Iblei and at the base of Mt. Etna. Normative mineralogy is summarized in Fig. 3. For Mt. Etna only basaltic lavas have been plotted, together with the main lavas emitted during the last 1971 eruption.

Two distinct groups of rocks (alkalic undersaturated and silica oversaturated tholeiites) are found at Mt. Iblei. The submarine base of Mt. Etna is probably mainly formed of olivine tholeiites. It is, however, to be pointed out that some nepheline-normative lavas occur among the submarine Etna products. The scattering of samples within the olivine-tholeiite field (Fig. 3) may partly reflect post-eruptive chemical changes in submarine lavas.

An interesting problem arises from the close spatial and temporal association of alkali basalts (and strongly undersaturated basanites) and tholeiites, which erupted in various episodes during the Plio–Pleistocene activity of Mt. Iblei. The lack of geochemical data hinders speculation as to whether the Iblei lavas have a common mantle source or not. They could have been generated by partial melting at different mantle depths or alternatively from the same mantle level with different degrees of partial melting [17]. A genetic linkage due to deep crystal fractionation is difficult to envisage owing to the strong bimodal compositional distribution of Mt. Iblei basalts (Fig. 3).

4. Paleomagnetic data

Some outcrops of volcanic rocks in eastern Sicily are particularly suitable for paleomagnetic study because they are emplaced upon a sufficiently stable carbonate platform. Samples of Upper Cretaceous Capo Passero dykes (8 sites) and of Plio–Pleistocene lavas from Mt. Iblei (10 sites) have been collected by core drilling (Tables 3 and 4). Details of field and laboratory techniques and analyses are described by Clark et al. [18]. The 80 m.y. old samples of Mt.

TABLE 3
Paleomagnetic data from Capo Passero site (36.7N, 15.1E)

Sample No.	N	D	I	K	α_{95}	P		dp	dm
						N	E		
Ib 4	3	175	−29	188	9.1	68	208	5.5	10.0
Ib 5	7	168	−38	16	15.0	71	232	10.0	18.1
Ib 6	6	172	−14	169	5.3	59	211	2.7	5.3
Ib 26	3	169	−25	384	6.2	64	220	3.8	6.9
Ib 27	4	167	−25	315	5.1	63	224	2.6	5.5
Ib 28	4	148	−27	46	13.6	53	255	8.1	14.4
Ib 29	4	160	−20	44	13.9	58	234	7.6	14.6
Ib 31	3	162	−25	311	4.8	61	234	2.8	5.2
Mean		165	−26	62	7.0	63	229	4.1	7.6
Mean*		167	−23	28	5.3	62.5	224	3.0	5.6

* Overall mean of paleomagnetic data from this paper and from Schult [19]. N = number of specimen, D = declination, I = inclination, K = Fisher precision parameter, α_{95} = circle of 95% confidence, P = virtual geomagnetic pole, dp and dm = polar error.

Tauro (4 sites) have been rejected owing to their high magnetic instability. Samples have been treated in a.c. fields up to 500 Oe peak values. Spurious components appeared to be removed after cleaning at 150–200 Oe. A.c. cleaning resulted in a more compact grouping of the sites without any major change in mean directions (Fig. 4). Reversal tests of Mt. Iblei lavas confirm that the samples are not affected by important spurious components.

The Upper Cretaceous pole evaluated for Sicily is consistent with that recently determined by Schult [19] on the same formation and with the African Upper Cretaceous pole [20]. These poles are significantly different (Fig. 5) from the mean Cretaceous pole for Europe estimated either from andesites from Bulgaria [21] or from andesites from England and Czechoslovakia [22].

The Plio–Pleistocene pole for Sicily as evaluated from the Mt. Iblei lavas (Fig. 5) is consistent [22] with the mean Upper Tertiary–Quaternary poles for the Central European–Russian platform and for France (Massif Central lavas) and with the Quaternary poles for Africa and Spain (Gerona volcanics).

The most important conclusion that can be drawn from the paleomagnetic data is that no significant differential motion of Sicily relative to Africa has occurred since the Upper Cretaceous. Sicily appears to have been integral with Africa in rotating anticlockwise, with a northwards shifting relative to Europe during the Tertiary. Furthermore, results from the Mt. Iblei volcanics suggest that, since the end of Miocene, Sicily has not undergone any noticeable movement relative to the stable Europe.

5. Discussion

Paleomagnetic data definitively indicate that at least since Upper Cretaceous time, Sicily has been part of the African plate, in agreement with the surface geology and with the continental nature of the crust in the Sicily Channel. It is reasonable to assume that during Mesozoic times also Sicily was integral with Africa.

Reconstructions of the relative movement of Africa with respect to Europe have recently been proposed on the basis of Atlantic spreading phases and least-squares fit of Mediterranean continental fragments [23]. These motions are consistent with paleomagnetic data available for Africa, Sicily and stable Europe (Fig. 5).

The volcanism of Sicily can now be utilized to check and integrate the geodynamic picture emerging from these studies.

TABLE 4
Paleomagnetic data from Mt. Iblei sites (37.2N, 14.8E)*

Sample	*N*	*D*	*I*	*K*	α_{95}	*P*		*dp*	*dm*
						N	E		
Ib 1	12	175	−54	88	4.6	85	261	4.6	6.5
Ib 2	6	185	−58	584	2.7	85	78	3.0	4.1
Ib 3	5	348	+11	11	28.9	57	216	14.9	29.3
Ib 14	11	180	−65	33	8.0	80	13	10.2	12.5
Ib 15	19	176	−48	148	2.8	81	219	2.3	3.6
Ib 16	7	178	−52	25	12.1	85	223	11.0	16.1
Ib 17	5	179	−47	37	12.0	81	205	10.3	16.0
Ib 19	5	1	+60	109	7.0	85	26	8.0	11.1
Ib 20	7	343	+40	66	7.1	69	244	5.0	8.9
Ib 023	7	12	+56	166	4.6	80	104	4.0	6.0
Mean		356	+48	24	10.0	81	219	9.1	13.9

*N = number of specimen, D = declination, I = inclination, K = Fisher precision parameter, α_{95} = circle of 95% confidence, P = virtual geomagnetic pole; dp and dm = polar error.

From the Upper Trias to the Upper Cretaceous, the African plate moved in a southeasterly direction with a sinistral shear movement with respect to Europe. The net result is a separation between the African and European plates, with the formation of oceanic crust.

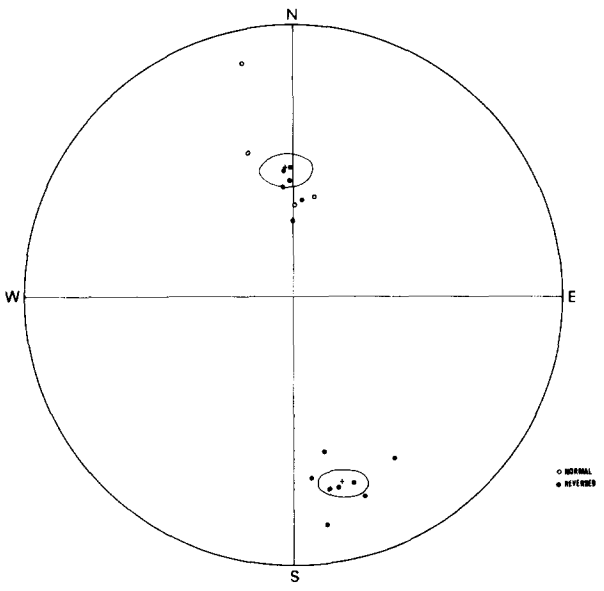
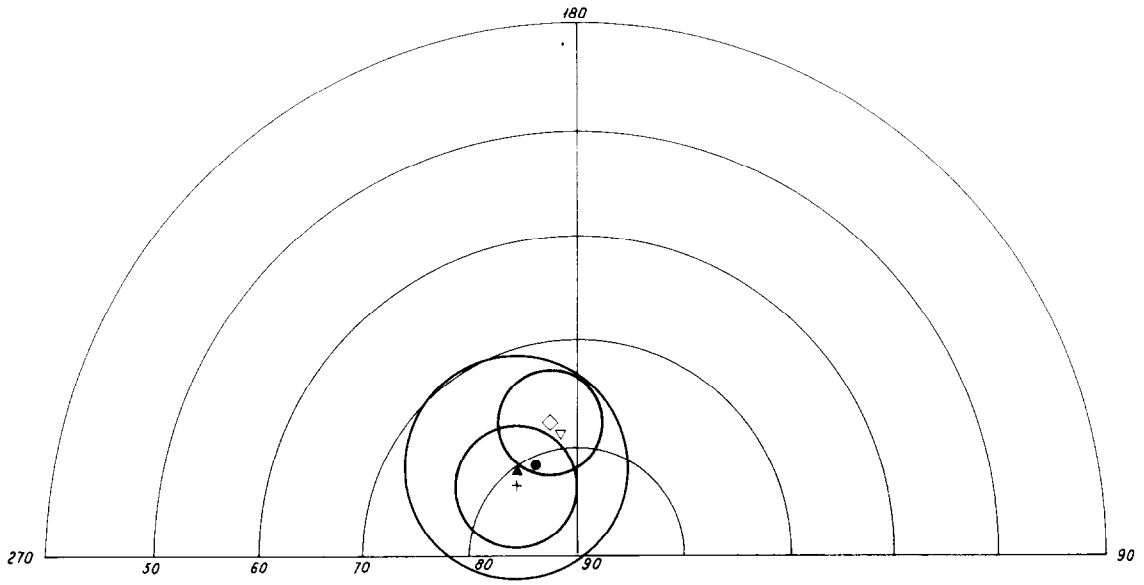


Fig. 4. Stereographic projection of mean sites directions of magnetization after a.c. cleaning of Mt. Iblei (Northern Hemisphere) and Capo Passero (Southern Hemisphere) samples. + = overall mean with 95% confidence circles.

The northern edge of the African continental plate (and, therefore, Sicily) was then affected by distensive tectonics. This is consistent with the nature of Sicilian volcanism at that time, which gave rise to alkalic products typical of continental distensive areas.

From Upper Cretaceous times to the present, as a consequence of the higher spreading rate of the north Atlantic, the motion of Africa with respect to Europe has been mainly compressional, with a minor dextral component. Volcanism is a sensitive marker of this important tectonic change. Basaltic distensive volcanism is in fact absent in eastern Sicily from the Upper Cretaceous to the Upper Miocene. The occurrence of an andesitic Upper Oligocene–Lower Miocene volcanism furthermore suggests that the compressional motion between Africa and Europe was accommodated by subduction of oceanic crust previously formed between Sicily and Europe. It produced an island-arc type volcanism toward the end of the Oligocene [8]. Oceanic crust consumption lasted until Lower Miocene,when the continental collision occurred, with the subsequent folding of the Peloritani range. Paleomagnetic data from Mt. Iblei indicate that in late Miocene times Sicily was already joined to Europe.

It must be pointed out that different episodes of continental collision occurred along the Sicilian–Calabrian segment of the plate boundary. In fact the subduction of oceanic crust beneath the Calabrian arc is still active, as indicated by the presence of the



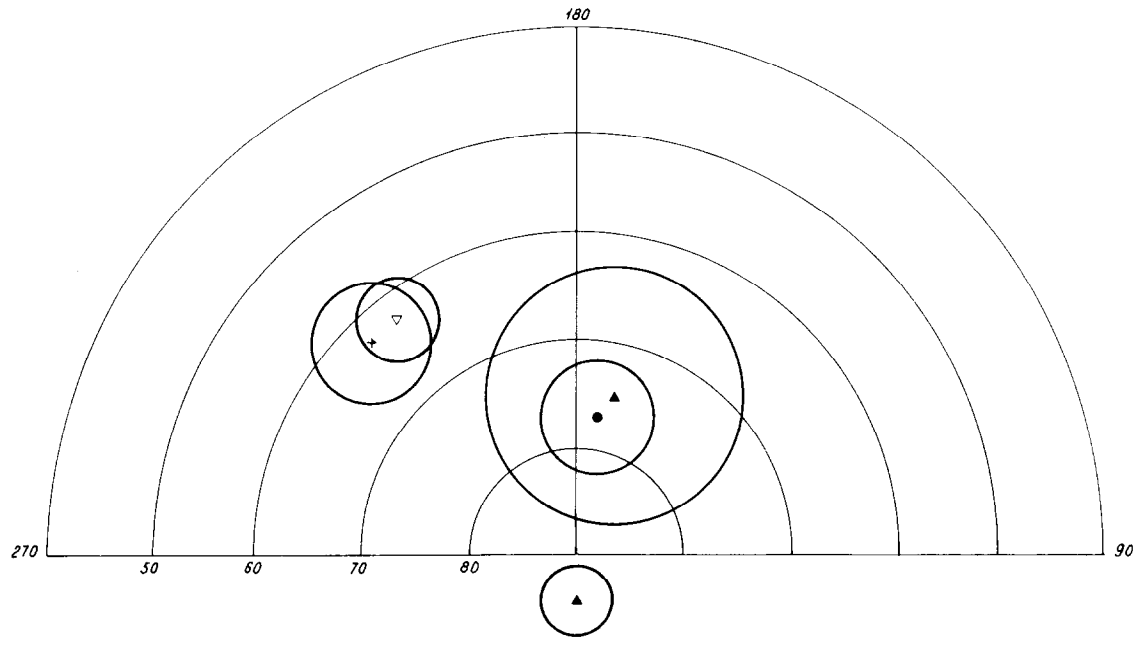


Fig. 5. (a) Upper Tertiary–Quaternary paleomagnetic poles with circles of confidence (when available); (b) Cretaceous and Eocene paleomagnetic poles with circles of confidence (when available).

Eolian island arc. On the other hand the continental nature of the crust on both sides of the boundary [24] suggests that consumption of oceanic crust is by now nearly complete.

Different times of continental collision along the plate boundary are probably related to the irregular shapes of the edges of the approaching continents.

If this interpretation is correct, it may also account for the occurrence since the Upper Miocene of basaltic distensive volcanism in eastern Sicily (Mt. Iblei and Mt. Etna), which would otherwise be difficult to explain within the general compressional tectonic framework. Since the Lower Miocene, eastern Sicily has in fact been the border zone between the colliding continental blocks and the still moving oceanic lithosphere subducting beneath the Calabrian arc. Therefore, eastern Sicily has been affected by local distensive tectonics related to the stress field directed normally to the motion of the oceanic lithosphere. The main consequence of this recent extensional tectonics was the formation of the Messina Abyssal Plain, a collapsed continental structure [24] located in the Ionian Sea southeast of Sicily (Fig. 1) and characterized by a strong Bouguer high (±300 mgal), suggesting a remarkable crustal thinning.

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